Worksheet 23 · Bounds IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Find the upper and lower bounds of a measurement • Calculate bounds of sums, differences, products and quotients • Use bounds to give an answer to an appropriate accuracy		
Grade: / 40	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Bounds*

Question 1 (4 marks)

The weight of a cat is 4.3 kg correct to 2 significant figures.

(a) Write down the upper bound of the weight of the cat.

..... kg
(1)

(b) Write down the lower bound of the weight of the cat.

..... kg
(1)

$G = e - f$

e = 17 correct to the nearest integer
f = 9.4 correct to one decimal place

(c) Work out the upper bound for the value of G.

.....
(2)

Question 2 (2 marks)

$C = b - a$

a = 6 correct to the nearest integer
b = 15 correct to the nearest 5

Work out the upper bound for the value of C
Show your working clearly.

Question 3 (4 marks)

$P = ef$

e = 4.8 correct to 2 significant figures.
f = 0.26 correct to 2 significant figures.

(a) Work out the lower bound for the value of P.
Show your working clearly.
Give your answer correct to 3 significant figures.

.....
(2)

$Q = \frac{t}{w}$

t = 2.73 correct to 3 significant figures.
w = 0.04 correct to 1 significant figure.

(b) Work out the upper bound for the value of Q.
Show your working clearly.
Give your answer correct to 2 significant figures.

(2)

Question 4 (3 marks)

$P = \frac{2a - c}{d}$

a = 58.4 correct to 3 significant figures.
c = 20 correct to 2 significant figures.
d = 3.6 correct to 2 significant figures.

Work out the upper bound for the value of P.
Show your working clearly.
Give your answer correct to 2 decimal places.

Question 5 (3 marks)

$$x = \frac{6a}{b - a}$$

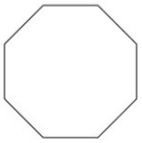
a = 3.46 correct to 3 significant figures.
b = 6.3 correct to 1 decimal place.

Work out the upper bound for the value of x.
Give your answer as a decimal correct to 3 significant figures.
Show your working clearly.

Question 6 (2 marks)

Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

Diagram **NOT** accurately drawn



(a) Write down the lower bound of the length of each side of the octagon.

..... mm
(1)

(b) Write down the upper bound of the length of each side of the octagon.

..... mm
(1)

Question 7 (3 marks)

$$a = \frac{v - u}{t}$$

v = 9.6 correct to 1 decimal place
u = 3.8 correct to 1 decimal place
t = 1.84 correct to 2 decimal places

Calculate the upper bound for the value of a.
Give your answer as a decimal correct to 2 decimal places.
Show your working clearly.

.....

Question 8 (4 marks)

A high speed train travels a distance of 487 km in 3 hours.

The distance is measured correct to the nearest kilometre.
The time is measured correct to the nearest minute.

By considering bounds, work out the average speed, in km/minute, of the train to a suitable degree of accuracy.
You must show all your working and give a reason for your answer.

Question 9 (3 marks)

$$X = \frac{2a - b}{f}$$

a = 7.5 correct to 1 decimal place.
b = 3.42 correct to 2 decimal places.
f = 2 correct to the nearest whole number.

Work out the upper bound of the value of X
Show your working clearly.

.....

Question 10 (3 marks)

$$A = w - \frac{x^2}{y}$$

w = 3.45 correct to 2 decimal places.
x = 1.9 correct to 1 decimal place.
y = 5 correct to the nearest whole number.

Work out the lower bound of the value of A
Show your working clearly.

.....

Question 11 (3 marks)

$$G = \frac{c}{2f - 3h}$$

c = 8 correct to the nearest whole number
f = 6.62 correct to 2 decimal places
h = 1.2 correct to 1 decimal place

Work out the lower bound for the value of G
Give your answer correct to 3 decimal places.
Show your working clearly.

.....

Question 12 (3 marks)

$$k = \frac{t}{a - h}$$

$t = 14$ correct to 2 significant figures
$a = 7.8$ correct to 2 significant figures
$h = 3.4$ correct to 2 significant figures

Work out the lower bound for the value of k.
Show your working clearly.

Question 13 (3 marks)

Here is a triangle XYZ.

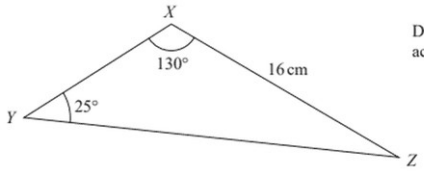


Diagram **NOT** accurately drawn

The length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.

Calculate the upper bound for the length YZ.
Give your answer correct to one decimal place.
Show your working clearly.